

Session 4: Train a Small GPT

From the GPT-2 Recipe to a Controlled Baseline

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Training Language Models: From First Principles to Efficient Serving

Lecture map

- 1 What a GPT-2-like model means
- 2 A controlled training baseline
- 3 Reading evidence from a run

Guiding question

What makes a small run a faithful experiment rather than merely a script that produces text?

What should be clear by the end?

- 1 Read a GPT configuration as shapes, parameters, memory, and work.
- 2 Connect the GPT-2 paper's architectural choices to nanoGPT's implementation.
- 3 Pre-register a baseline whose changes can be interpreted.
- 4 Use curves, samples, and checkpoints as different kinds of evidence.

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GPT-2 tested a simple scaling hypothesis

Core idea

Train one autoregressive objective on diverse web text. Capabilities may emerge from scale without task-specific training.

$$\max_{\theta} \sum_t \log p_{\theta}(x_t \mid x_{<t})$$

- The objective stays fixed across examples and tasks.
- Model and data scale change, not the loss definition.
- Evaluation asks what transfers from next-token prediction.

Source: Radford et al., Language Models are Unsupervised Multitask Learners

GPT-2-like names an architectural recipe

Retained in a small baseline

- decoder-only causal attention
- pre-norm residual blocks
- GELU MLP expansion
- learned absolute positions
- tied token embedding and LM head

Reduced for teaching

- fewer layers
- narrower residual stream
- shorter context
- smaller data and token budget

The mechanism remains recognizable even when the scale changes.

Configuration is a compact model of the run

Field	Structural effect	Systems effect
depth L	repeated computation	activations, sequential depth
width d	residual representation	parameters scale roughly as d^2
heads H	attention subspaces	requires $d \bmod H = 0$
context T	visible history	attention scores scale as T^2
vocabulary V	output choices	embedding and softmax size
micro-batch B	examples per pass	activation memory

Before launch

Derive $d_h = d/H$, tokens per update, approximate parameter count, and expected peak memory.

Most block parameters come from six matrices

For a GPT-2-style block with MLP width $4d$:

$$\underbrace{3d^2 + d^2}_{\text{QKV and output}} + \underbrace{4d^2 + 4d^2}_{\text{MLP}} \approx 12d^2$$

$$N_{\text{params}} \approx 12Ld^2 + Vd + Td$$

Worked estimate

$L = 12$, $d = 768$, $V \approx 50\text{k}$ gives about 124M parameters before small bias and norm terms, close to GPT-2 small.

Why estimate first?

A surprising exact count often reveals untied embeddings, a wrong MLP width, or an accidental extra projection.

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Initialization must respect repeated residual additions

$$X_{\ell+1} = X_{\ell} + F_{\ell}(X_{\ell})$$

If every branch contributes similar variance, depth accumulates it.
GPT-style implementations commonly scale residual output projections:

$$\sigma_{\text{proj}} = \frac{0.02}{\sqrt{2L}}$$

Ordinary weights

Initialize with standard deviation near 0.02.

Residual outputs

Scale attention and MLP output projections by depth.

Source: GPT-2 implementation convention reproduced in nanoGPT

Weight tying changes capacity and optimization

$$W_{\text{lm-head}} = W_E$$

- The same token geometry supports input lookup and output scoring.
- Parameter count falls by Vd .
- Both input and output losses update one shared matrix.

Audit

Confirm shared storage, not merely equal initial values. Two copied matrices will diverge after the first update.

The experiment contract exists before the first batch

- 1 Freeze data version, tokenizer, split, and sampling policy.
- 2 Record model configuration and initialization convention.
- 3 Fix tokens per update, optimizer, schedule, and total token budget.
- 4 Declare evaluation cadence, validation token count, and seeds.
- 5 Name the selection rule for the final checkpoint.

Controlled comparison

Change one explanatory variable while holding the token and evaluation budgets fixed.

Tokens, not epochs, are the common unit

$$N_{\text{update}} = B_{\text{micro}} T K W$$

$$N_{\text{seen}} = N_{\text{update}} \times \text{optimizer steps}$$

Fair run comparison

Match tokens seen and validation tokens evaluated.

Misleading comparison

Matching wall time can reward a faster system with more optimization work.

Checkpoint: pre-registration

Think, then compare

You want to test a new attention kernel. Which fields must remain fixed to support a quality claim, and which additional measurements support a speed claim?

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Four curves tell different stories

Optimization

- training loss
- gradient norm
- learning rate

Generalization and systems

- validation loss
- tokens per second
- peak allocated memory

Interpret jointly

A loss spike aligned with a learning-rate transition suggests optimization. A throughput drop with stable loss suggests the system path. A widening train-validation gap suggests overfitting or distribution shift.

Perplexity is an exponentiated token-level loss

$$\text{PPL} = \exp(\mathcal{L}_{\text{CE}})$$

- It is interpretable only for the same tokenizer and evaluation distribution.
- Small cross-entropy changes can look larger after exponentiation.
- Report the underlying mean loss and token count with perplexity.

Not a universal model score

Different tokenizers create different prediction units, so their perplexities are not directly comparable.

Samples are diagnostics, not the selection metric

Fix prompts and sampling settings across checkpoints.

Useful for detecting

- repetition loops
- broken boundaries
- memorized fragments
- loss of coherence

Poor evidence for

- small quality differences
- cross-run ranking
- claims without fixed prompts and seeds

Checkpoint choice answers a declared question

- 1 **Latest**: exact continuation and operational recovery.
- 2 **Best validation loss**: model selection under a fixed protocol.
- 3 **Fixed token budget**: fair comparison across experimental arms.

Resume integrity

Restore optimizer, scaler, step, sampler, and RNG state. Verify the next update against an uninterrupted run.

Checkpoint: curve triage

Think, then compare

Training loss improves, validation loss stalls, samples become more fluent, and throughput is unchanged. Which conclusions are supported, and what experiment should come next?

The durable ideas

- 1 A small GPT is faithful when it preserves the architectural mechanism and states its reductions.
- 2 Configuration fields imply tensor shapes, parameter count, memory, and compute.
- 3 A baseline is defined by a frozen experimental contract, not by one successful run.
- 4 Curves, perplexity, samples, and checkpoints answer different questions.